Trudy before it exists: what the five models that got Simon right say for Puerto Vallarta

The five best position forecasters of Simon's last two days, leaving out the NHC's own products: Google DeepMind's cyclone model, the ECMWF AIFS, the track consensus, the Canadian ensemble mean and the GEFS ensemble mean. Their newest lines for EP93, how they moved in six hours, the chances they imply for Vallarta, and the least, most likely and worst cases at D Terrace in Amapas, Casa Francisca in Emiliano Zapata and the house in El Pitillal.

Monday 12 October 2026, 7 pm Vallarta time (13 Oct 01Z) · the five models' Monday 18Z runs and their 00Z early versions · NHC outlook of 5 pm PDT · no NHC advisory exists yet for EP93 · not an official forecast

Bottom line

- **The newest numbers moved away from the bay.** At 6 pm the NHC placed EP93's center at 8.3N 95.3W, 156 km west of its noon position and roughly 100 km west of where the five models expected it. Moved to that center (the "early versions" the NHC forecaster works from), the five pass ECMWF AIFS 168 km WNW, Google DeepMind cyclone model 216 km WSW, track consensus TVCN 289 km SW, Canadian ensemble mean 482 km SSW, GEFS ensemble mean 650 km SSW. Six hours earlier, three of the five passed within 121 km.
- **The chances, read from the five.** Newest versions: center within 300 km, **60%** (three of the five); within 200 km, **20%**; within 100 km, **0%**; tropical-storm wind in the city, **0%**. The 18Z runs gave 60%, 60%, 20% and 60%. The ECMWF AIFS ensemble (18Z) still puts 31% of its members within 100 km, the ECMWF ensemble (12Z) 22%, the GEFS 0%, climatology 9%.
- **Our reading.** The center of a broad low is uncertain by 100 km or more until it tightens, so one re-fix should not erase the earlier runs. Weighting the newest positioning most: **about 2 in 3** that the center passes within 300 km and brings swell and rain bands, **about 1 in 5** for tropical-storm wind in the city, **about 1 in 10** for the center within 100 km.
- **When.** Sunday 18 October from midday to Monday 19 morning. Closest approach on the newest lines: Sunday 6 pm (DeepMind, AIFS) to Monday 3 am (track consensus). Swell from Saturday; rain bands from Sunday midday.
- **Most likely** (the median of the five, the track consensus TVCN 00Z): center 289 km SW at Mon 19 Oct 03:00 as a category-1 hurricane; rain bands Sun 18 Oct noon to Mon 19 Oct noon; 26 km/h with gusts to 37 on the Amapas terrace; about 31 mm. **Worst case** (the Google DeepMind cyclone model 18Z, before the re-fix): center 110 km W at Sun 18 Oct 17:00 as a category-1 hurricane; tropical-storm wind Sun 18 Oct 10:00 to Sun 18 Oct 20:00; 65 km/h with gusts to 90 at Amapas; about 123 mm before the mountains add theirs. **Least** (the GEFS ensemble mean 00Z): a weak storm 650 km to the south; nothing here.
- **Outside the five.** The NHC corrected consensus, excluded here as an NHC product, moved the other way: its 00Z version passes 67 km W at Sun 18 Oct 21:00 after a 141-kt peak. The ECMWF IFS 12Z run makes landfall on Bahía de Banderas. The GFS, HAFS and HWRF send it far out to sea.
- **What to do now:** nothing physical. Keep Sunday flexible. NHC advisory 1 is expected tonight or Tuesday; its wind-speed probabilities for Puerto Vallarta replace every number here. Thursday noon, 72 hours out, is the decision point.

60%
center within 300 km: three of the five, newest versions (60% six hours earlier)

20%
center within 200 km: one of the five (60% six hours earlier)

0%
tropical-storm wind in the city on the newest five (60% six hours earlier); our reading about 1 in 5

Sun–Mon
closest approach Sunday 6 pm to Monday 3 am on the newest lines

1. Where EP93 is and what the NHC says

NHC best-track fixes on Monday: 7.9N 93.1W at 6 am, 8.1N 93.9W at noon, 8.3N 95.3W at 6 pm, with 25-kt winds and 1008 mb at the last fix, about 1,700 km southeast of Puerto Vallarta. The 156-km move between noon and 6 pm is faster than the motion the outlook describes (about 16 km/h, or 100 km in six hours), which points to the broad center being re-fixed farther west rather than a real jump. The 5 pm PDT outlook: "Showers and thunderstorms are beginning to show signs of organization in association with a broad area of low pressure located several hundred miles south-southwest of the coast of Guatemala. This system is gradually becoming better defined, and environmental conditions are forecast to be conducive for further development. A tropical depression or tropical storm is expected to form well south of southern Mexico later tonight or tomorrow while the system moves generally west-northwestward at around 10 mph." Formation chance through 48 hours and 7 days: near 100 percent. The next name on the 2026 list is Trudy.

2. The five, and how they moved

The final Simon report scored every model's position forecasts against the NHC best track over the two days before the pass (mean error at 12, 24 and 36 h, its Figure 2). Leaving out the NHC's own two products (the official forecast, 47 km, and the corrected consensus, 38 km) and the climatology-persistence baseline (53 km, a yardstick rather than a model), the five best were:

Model	Simon km	Weight	00Z: closest km	Side	When	kt then	Peak kt	18Z: closest	Amapas km/h	E. Zapata	Pitillal	Rain mm
Google DeepMind cyclone model	33	39%	**216**	WSW	Sun 18 18:00	79	121	110 WSW	38	33	29	65
ECMWF AIFS †	50	17%	**168**	WNW	Sun 18 18:00	68	94	82 WNW	47	41	36	36
Track consensus TVCN †	51	16%	**289**	SW	Mon 19 03:00	64	94	121 SW	26	23	20	31
Canadian ensemble mean	54	15%	**482**	SSW	Sun 18 08:00	40	41	426 SSW	11	9	8	14
GEFS ensemble mean	57	13%	**650**	SSW	Sun 18 18:00	39	49	604 SSW	8	7	6	3

00Z columns: the early versions. "Simon km" is the 12–36 h mean position error; the weight is its inverse square, normalised over the five. Closest approach to 20.61N 105.23W, Vallarta time. Building winds are sustained, on the 00Z line. † intensity from the NHC intensity consensus IVCN.

- **What an early version is.** The NHC cannot wait for a run to finish, so at each cycle it takes the previous run and moves it so that its 6-hour position matches the new center, with the same shift at every hour. The 00Z DeepMind, track consensus and GEFS-mean lines here are the NHC's own; the Canadian mean is a 12-hour-old run moved the same way. No early AIFS exists in the NHC files, so we built one by the same rule from the 18Z AIFS run; the shift was 0.5° south and 1.0° west.
- **Two caveats about the five.** The track consensus TVCN is computed at the NHC but is a plain average of the GFS, ECMWF, UKMET, HWRF, HMON, HAFS and COAMPS tracks, so it is kept; Figure 4 shows what changes if the corrected consensus is let back in. The AIFS, like every gridded AI model, under-forecasts intensity by construction, and the TVCN has no intensity at all; both carry the NHC intensity consensus IVCN, which is how the NHC uses TVCN.
- **What Simon's scores can and cannot say.** They were earned at 12–36 h on a slow, stalling storm near the coast. Nothing guarantees they transfer to a six-day forecast of a storm that has not formed. Five lines cannot resolve a probability finer than 20%; the ensembles and the climatology are the context that keeps the five honest.
- **Center winds are not Vallarta's winds.** Simon reached category 4 and passed 120 km away; the airport recorded 7 km/h and under 10 mm. The building numbers say "if the center followed that line"; once they exist, the NHC wind-speed probabilities are the number to use.

3. The lines on the map

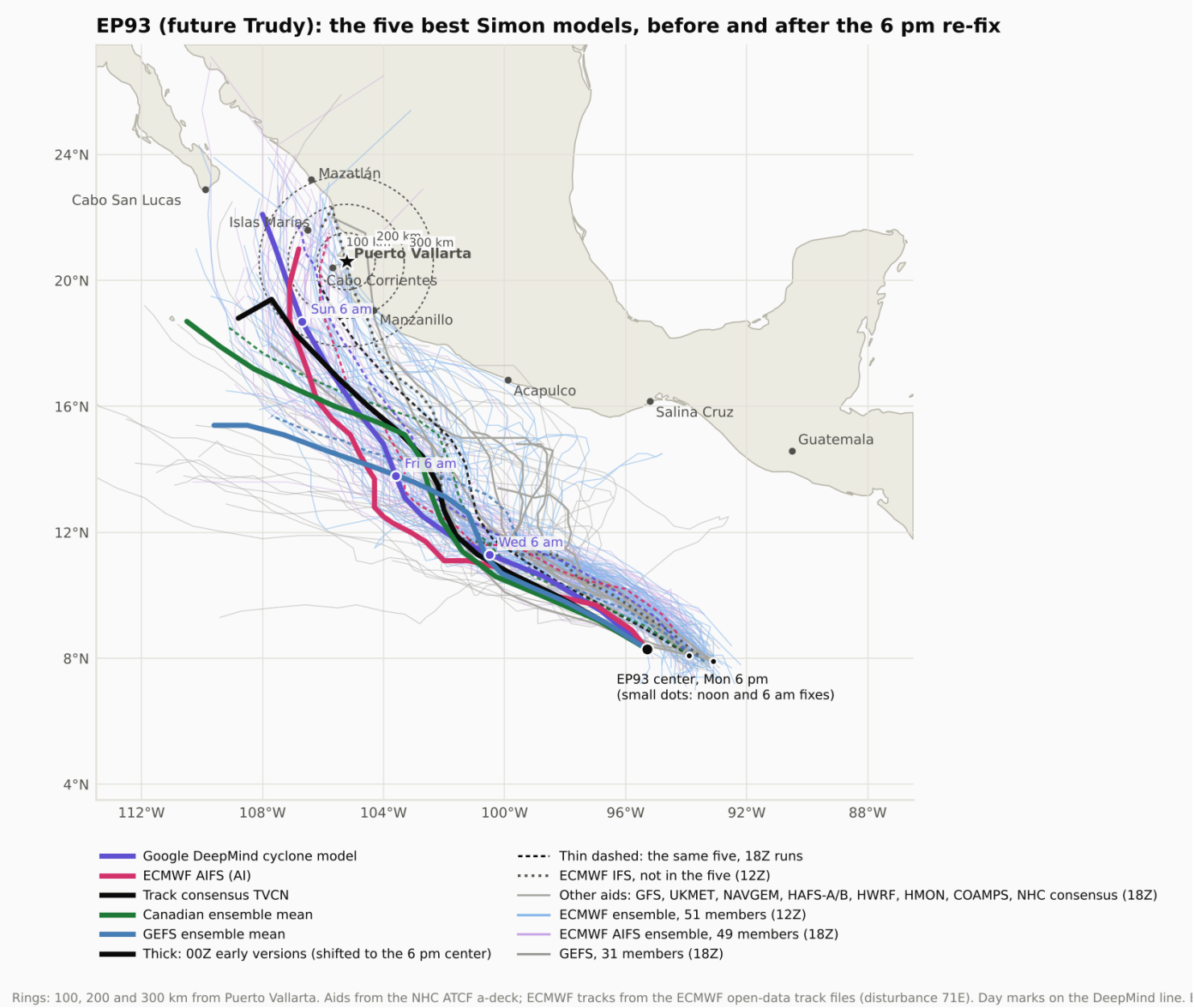


Figure 1. The five best Simon models: thick, the 00Z early versions; thin dashed, the same five from the 18Z runs. The whole bundle moved about 100 km west with the re-fixed center. Grey: the other aids (18Z); faint: the three ensembles.

The five still split the way all the guidance splits. The GEFS mean and the Canadian mean keep EP93 a weak system heading west; DeepMind, the AIFS and the track consensus make it a hurricane that turns north toward Cabo Corrientes from Friday, then pass it west or southwest of the bay on Sunday. Deeper storms are steered by deeper flow, so intensity and track are one question here. The ECMWF ensemble (12Z) puts 53% of 51 members within 300 km and 22% within 100 km; the AIFS ensemble (18Z) 78% and 31%; the GEFS (18Z) 6% and 0%. None of the three has been moved to the 6 pm center.

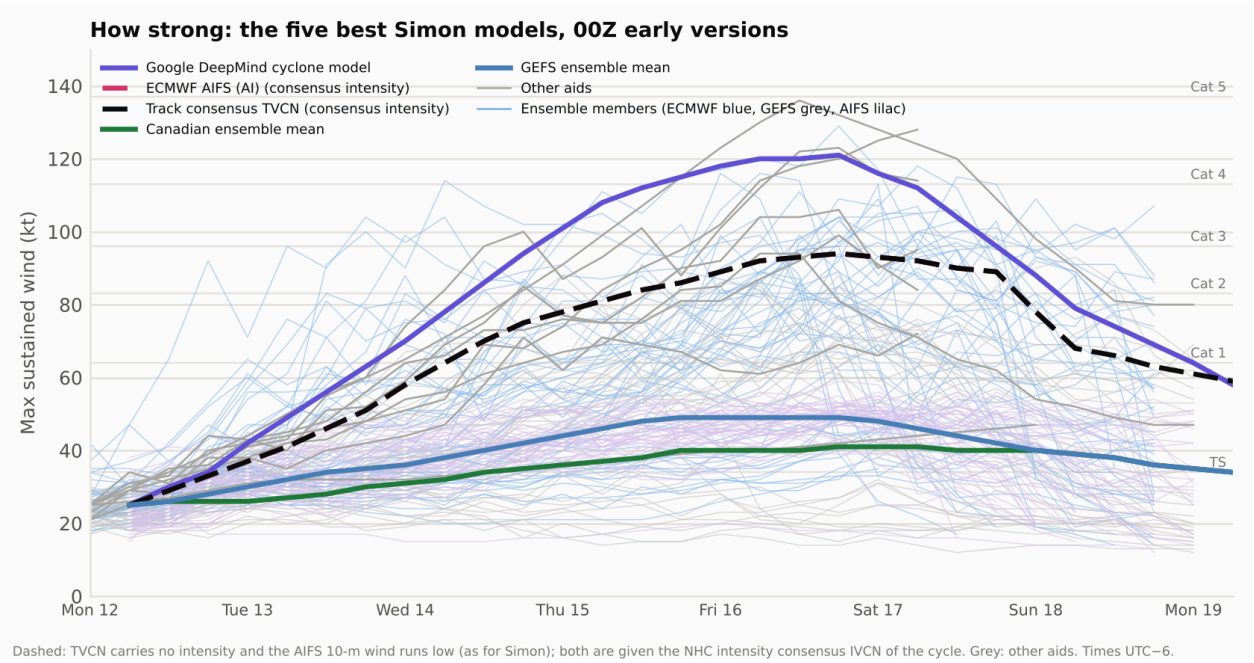


Figure 2. Intensity on the five newest lines. DeepMind peaks at 121 kt; the intensity consensus carried by the AIFS and the track consensus peaks at 94 kt; the two ensemble means never reach hurricane strength.

4. Climatology, for context

Of the 33 storms since 1949 that passed within 500 km of EP93's 6 pm position between 12 September and 11 November, 11 came within 300 km of Puerto Vallarta, 6 within 200 km and 3 within 100 km. 22 became hurricanes and 14 major hurricanes. The closest passes came 2.8–6.5 days after the analog point (one outlier at 16 days), which from Monday evening means Thursday to Monday. Climatology alone puts a 300-km pass at 33% and a 100-km pass at 9%.

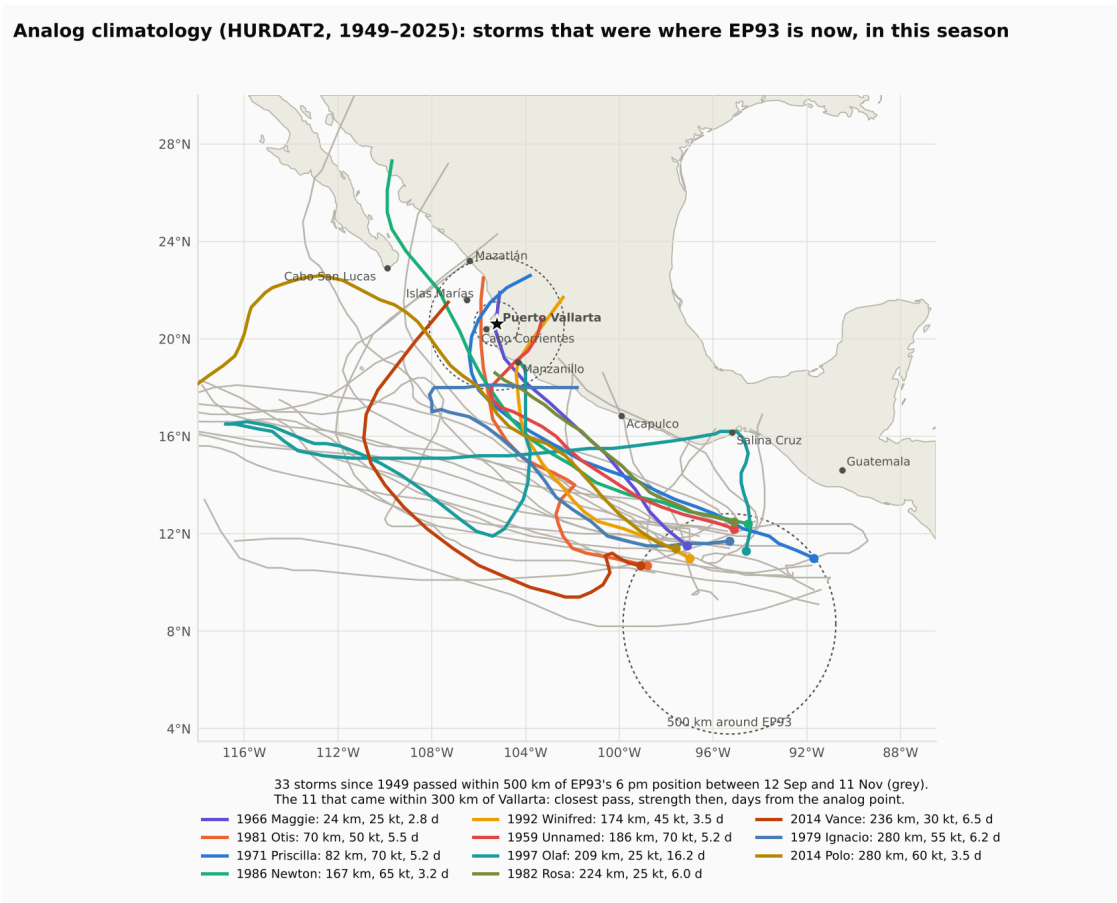


Figure 3. The 33 analog storms (grey) and the 11 that came within 300 km of Vallarta.

Storm	Analog point	Closest, km	When	kt then	Peak kt	Region of closest pass
1966 Maggie	11.5N 97.1W, 1966-10-16	24	1966-10-19 (+2.8 d)	25	45	Jalisco (Bahía de Banderas)
1981 Otis	10.7N 98.8W, 1981-10-24	70	1981-10-29 (+5.5 d)	50	75	Jalisco (Bahía de Banderas)
1971 Priscilla	11.0N 91.7W, 1971-10-07	82	1971-10-12 (+5.2 d)	70	110	Nayarit
1986 Newton	12.4N 94.5W, 1986-09-18	167	1986-09-21 (+3.2 d)	65	75	Sinaloa
1992 Winifred	11.0N 97.0W, 1992-10-06	174	1992-10-10 (+3.5 d)	45	100	Michoacán / Colima
1959 Unnamed	12.2N 95.1W, 1959-10-22	186	1959-10-27 (+5.2 d)	70	120	Michoacán / Colima
1997 Olaf	11.3N 94.6W, 1997-09-26	209	1997-10-12 (+16.2 d)	25	60	Michoacán / Colima
1982 Rosa	12.5N 95.1W, 1982-09-30	224	1982-10-06 (+6.0 d)	25	45	Jalisco (Bahía de Banderas)
2014 Vance	10.7N 99.1W, 2014-10-29	236	2014-11-05 (+6.5 d)	30	95	Sinaloa
1979 Ignacio	11.7N 95.3W, 1979-10-23	280	1979-10-30 (+6.2 d)	55	125	Jalisco (Bahía de Banderas)
2014 Polo	11.4N 97.6W, 2014-09-16	280	2014-09-19 (+3.5 d)	60	65	Sinaloa

5. The chances

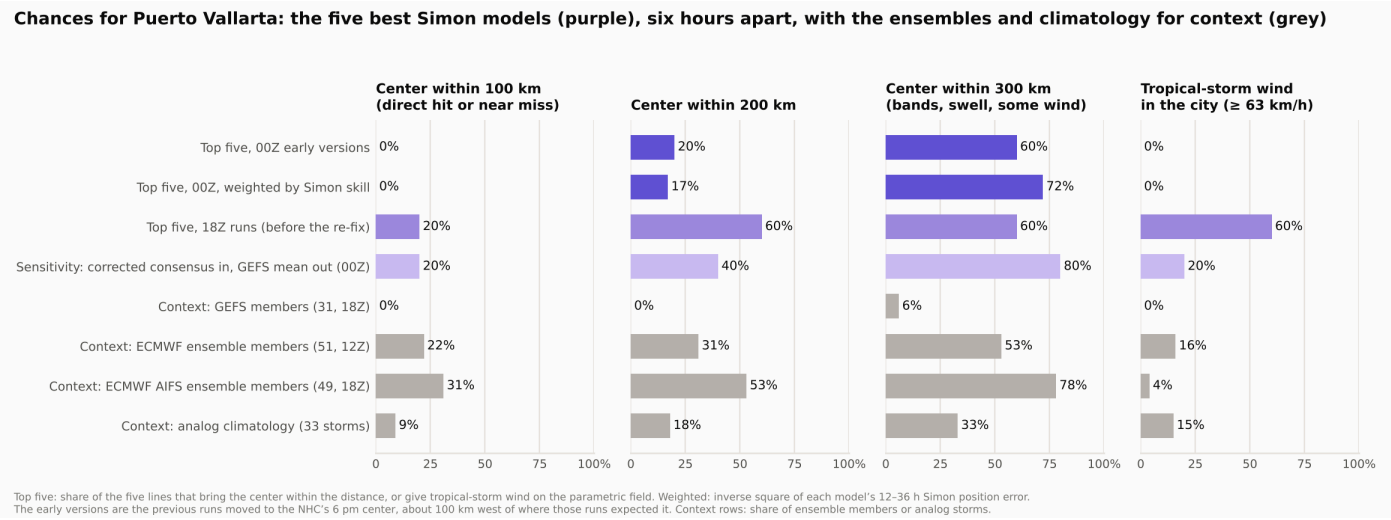


Figure 4. Share of the five lines that bring the center within 100, 200 and 300 km of Puerto Vallarta and that give tropical-storm wind in the city, newest versions (dark), the 18Z runs (mid), and with the corrected consensus swapped in for the GEFS mean (light). Grey: ensembles and climatology.

Outcome	Five, 00Z	Weighted	Five, 18Z	AIFS ens. / ECMWF ens. / GEFS / climo	Our reading
Center within 300 km: swell, rain bands, gusty showers	60%	72%	60%	78 / 53 / 6 / 33%	about 2 in 3
Center within 200 km: heavy rain likely, tropical-storm gusts possible on the exposed shore	20%	17%	60%	53 / 31 / 0 / 18%	about 1 in 3
Center within 100 km: the hurricane itself could reach the bay	0%	0%	20%	31 / 22 / 0 / 9%	about 1 in 10
Tropical-storm wind (≥ 63 km/h sustained) in the city	0%	0%	60%	4 / 16 / 0 / 15%	about 1 in 5
Hurricane wind (≥ 119 km/h sustained) in the city	0%	0%	0%	0 / 0 / 0 / –	under 5%

How the reading is made. The newest five give no tropical-storm wind and no 100-km pass. The same five six hours earlier gave three of five for both kinds of close outcome, and the only thing that changed was where the NHC placed a broad, still-forming center. The AIFS ensemble, the best of the three ensembles on Simon's track, still has about a third of its members within 100 km. Climatology sits near 1 in 10 for a 100-km pass. Our reading weights the newest positioning most but does not discard the rest. It is a judgment, rounded on purpose, and it will move with the first NHC advisory.

6. The three addresses: least, most likely, worst

Each column is one of the five lines, run through the same parametric wind field and exposure factors as the Simon graphics (1.08 for the Amapas terrace when the wind comes off the bay, 0.89 otherwise; 0.77 for the sheltered mid floor; 0.70 for the street-level house), with an R-CLIPER-type rain total along the track (no terrain enhancement; the Sierra behind the city can double the total on the slopes). "Most likely" is the median of the five newest lines by closest approach. "Worst" is the line, from either the newest or the 18Z set, with the strongest wind at the terrace among lines that carry their own intensity. Times are Vallarta time.

	Least impact **GEFS ensemble mean 00Z: 650 km S at Sun 18 Oct 18:00, 39 kt**	**Most likely** **track consensus TVCN 00Z: center 289 km SW at Mon 19 Oct 03:00, category-1 hurricane (64 kt, consensus intensity)**	**Worst case** **Google DeepMind cyclone model 18Z: center 110 km W at Sun 18 Oct 17:00, category-1 hurricane (80 kt) after a 125-kt peak**
Swell on the coast	–	Sat 17 Oct midnight → Tue 20 Oct midnight	Fri 16 Oct 18:00 → Mon 19 Oct 18:00
Rain bands over the bay	–	Sun 18 Oct noon → Mon 19 Oct noon	Sat 17 Oct 18:00 → Mon 19 Oct 18:00
Tropical-storm wind in the city	–	–	Sun 18 Oct 10:00 → Sun 18 Oct 20:00
Rain, storm total (city, before terrain)	3 mm	31 mm	123 mm
D Terrace, Amapas, 8th floor	**8** km/h, gusts 11 Sun 18 Oct 04:00, wind from the ENE	**26** km/h, gusts 37 Mon 19 Oct midnight, wind from the ESE	**65** km/h, gusts 90 Sun 18 Oct 15:00, wind from the ESE
Casa Francisca, Emiliano Zapata, 3rd floor	**7** km/h, gusts 10 Sun 18 Oct 04:00, wind from the ENE	**23** km/h, gusts 33 Mon 19 Oct midnight, wind from the ESE	**56** km/h, gusts 81 Sun 18 Oct 15:00, wind from the ESE
El Pitillal house, street level	**6** km/h, gusts 9 Sun 18 Oct 03:00, wind from the ENE	**20** km/h, gusts 29 Mon 19 Oct 01:00, wind from the ESE	**49** km/h, gusts 72 Sun 18 Oct 15:00, wind from the ESE
What it means	A weak storm far to the south. Normal surf; nothing else.	A hurricane passing well offshore to the southwest overnight Sunday, with the city on its onshore side. Big surf on the open coast and the Malecón from Saturday; a wet, gusty Sunday afternoon and night; balcony items at Amapas worth bringing in. Casa Francisca and Pitillal: ordinary heavy-rain precautions, keep the car off the low streets near the river on Sunday night.	A category-1 hurricane about 110 km west of the bay on Sunday afternoon, moving north. Tropical-storm wind for most of Sunday, gusts to 90 on the Amapas terrace, power cuts likely; 123 mm plus terrain on the Cuale and Pitillal catchments puts both rivers up and the low streets under water, with slides possible on the Amapas and Conchas Chinas slopes. Pitillal: be on higher ground with the car by Saturday night. A Hurricane Watch on Friday would be about this case.

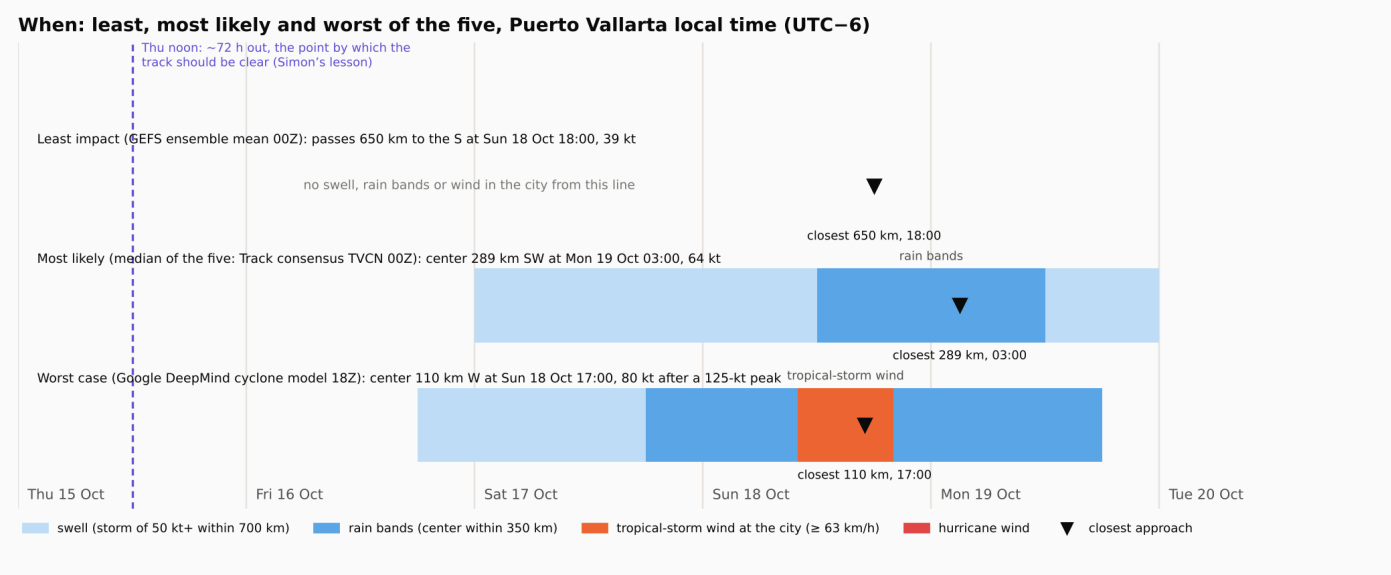


Figure 5. The three cases in time. The triangle is the closest approach of the center.

7. When we will know

- **Tonight or Tuesday:** NHC advisory 1 on Tropical Depression or Tropical Storm Trudy, with the first official five-day line and wind-speed probabilities. If Puerto Vallarta appears in the 34-kt probability table at all (the threshold is 3%), the close cases are alive in the official forecast.
- **Tuesday morning and evening:** the 00Z and 12Z model runs, the first that start from the re-fixed center rather than being moved to it. If the five come back east toward the bay, the 6 pm re-fix was the outlier; if they stay west, the most-likely case firms up.
- **Thursday noon, about 72 hours out:** the decision point. At three days the NHC's average track error is about 150 km. If the official line then runs within 150 km of the bay, prepare for the worst case; if it runs 250 km or more to the southwest, plan for swell and rain. Simon's lesson is that even inside 24 hours the line can shift 100 km; read the probabilities, not the line.
- **Friday:** the SMN and the NHC would issue a Hurricane Watch for the Jalisco coast about 48 hours before tropical-storm winds are expected, if the close cases are still on the table.

8. Simon, meanwhile

NHC advisory 22A (Monday 6 pm Vallarta time): Simon is a category-2 hurricane with 85-kt winds about 125 km west of Cabo Corrientes, drifting west-northwest at 6 km/h. It is forecast to move slowly away from the bay, then turn northeast and reach the Sinaloa coast as a weakening hurricane around Thursday night or Friday. A Hurricane Warning is in effect for the Islas Marías and a Hurricane Watch from Cabo Corrientes to Bahía Tempehuaya. Puerto Vallarta's wind probabilities are below the 3% display threshold; Mazatlán's 34-kt cumulative probability is 54%. Simon will be inland over Sinaloa before Trudy's window opens; the swell it leaves on the Jalisco coast (4–5 m per SMN aviso 40) should ease by midweek, before Trudy's own swell arrives.

9. Method, sources and caveats

- **The five:** Google DeepMind cyclone model (ATCF aids GDMN 18Z, GDMI 00Z), ECMWF AIFS single run (ECMWF open-data tropical-cyclone track files, 18Z, disturbance 71E, decoded with ecCodes; the 00Z early version built by the NHC rule), track consensus TVCN, Canadian ensemble mean (CEMI 18Z, CEM2 00Z), GEFS ensemble mean (AEMN 18Z, AEMI 00Z).
- **Intensity:** each model's own, except the TVCN (none) and the AIFS (model 10-m wind, low by construction), which carry the NHC intensity consensus IVCN of their cycle, interpolated to each hour.
- **Closest approach, timing, landfall, building winds, rain:** every track interpolated hourly; distance and timing to 20.61N 105.23W; landfall is the first hour over land in the Natural Earth 50 m polygons; wind from the parametric field of the Simon map page with the model's own 34/50/64-kt radii where the deck carries them and climatological radii otherwise, times the Simon exposure factors, with no decay over land; rain from an R-CLIPER-type profile (after Tuleya et al. 2007) integrated hourly, with no terrain enhancement.
- **Weights:** inverse square of each model's 12–36 h mean position error over Simon's last 48 h (final report, Figure 2): DeepMind 33, AIFS 50, TVCN 51, Canadian ensemble mean 54, GEFS mean 57 km.
- **Context:** GEFS members and all other aids from the NHC ATCF a-deck (aep932026.dat, as of 01Z on 13 October); best track bep932026.dat; ECMWF IFS ensemble (12Z, 51 members) and AIFS ensemble (18Z) from the ECMWF open-data track files (the 18Z IFS files stop at 144 h, short of the window); HURDAT2 NE Pacific 1949–2025 for the analogs, centred on the 6 pm position.
- **Not included:** the real 00Z model runs (not yet published); any NHC advisory, because none exists. This document should be rerun when advisory 1 and the Tuesday runs are out.

Appendix. Every aid

A1. 00Z early versions (what the NHC forecaster sees now)

Aid	Closest, km	Side	When (local)	kt then	Peak kt	Amapas km/h	E. Zapata	Pitillal	Rain mm	Landfall
Google DeepMind (early)	216	WSW	Sun 18 18:00	79	121	38	33	29	65	–
ECMWF AIFS (AI, early) †	168	WNW	Sun 18 18:00	68	94	47	41	36	36	–
Track consensus †	289	SW	Mon 19 03:00	64	94	26	23	20	31	–
Canadian ens. mean (early, 12 h)	482	SSW	Sun 18 08:00	40	41	11	9	8	14	–
GEFS mean (early)	650	SSW	Sun 18 18:00	39	49	8	7	6	3	–
NHC corrected consensus	67	W	Sun 18 21:00	84	141	84	70	63	203	–
GFS (early)	551	S	Mon 19 18:00	50	89	10	9	8	3	–
UKMET (early, 12 h)	433	SW	Sun 18 08:00	45	45	20	17	16	11	–
NAVGEM (early, 12 h)	339	SW	Sat 17 16:00	44	97	20	17	15	27	Baja California Sur, Mon 19 16:00, 22 kt
Canadian GEM (early, 12 h)	412	WSW	Sun 18 13:00	58	59	34	29	26	20	–
HAFS-A (early)	626	SSE	Sun 18 06:00	135	135	18	15	14	0	–
HAFS-B (early)	793	SSE	Sun 18 06:00	89	100	11	9	8	0	–
HWRF (early)	807	SSE	Sun 18 06:00	77	77	13	11	10	0	–
HMON (early)	235	SSW	Sun 18 06:00	75	92	23	20	18	17	–
COAMPS-TC (early)	358	SSW	Sun 18 06:00	108	119	29	25	22	5	–
Climatology-persistence	680	SSW	Sat 17 01:00	–	–	0	0	0	0	–

A2. 18Z runs (late versions where available)

Aid	Closest, km	Side	When (local)	kt then	Peak kt	Amapas km/h	E. Zapata	Pitillal	Rain mm	Landfall
Google DeepMind	110	WSW	Sun 18 17:00	80	125	65	56	49	123	–
ECMWF AIFS (AI) †	82	WNW	Sun 18 14:00	81	88	81	66	59	86	–
Track consensus †	121	SW	Mon 19 noon	69	88	62	53	47	74	–
Canadian ens. mean (early)	426	SSW	Sun 18 16:00	40	42	12	11	10	19	–
GEFS mean	604	SSW	Mon 19 09:00	34	49	8	7	6	4	–
NHC corrected consensus	74	E	Sun 18 15:00	94	136	100	71	65	178	Michoacán / Colima, Sun 18 02:00, 116 kt
GFS	603	SSE	Mon 19 noon	47	94	10	9	8	2	–
UKMET (early)	397	SW	Sun 18 noon	47	47	22	19	17	10	–
HAFS-A	744	SE	Sat 17 18:00	128	128	14	12	11	0	–
HAFS-B	879	SE	Sat 17 18:00	95	106	10	8	8	0	–
HWRF	964	SE	Sat 17 18:00	72	72	13	11	10	0	–
HMON	396	SSE	Sat 17 18:00	84	99	18	16	14	2	–
COAMPS-TC	453	S	Sat 17 18:00	114	123	27	24	21	1	–
Climatology-persistence	540	SSW	Sat 17 noon	–	–	0	0	0	0	–

† intensity from the NHC intensity consensus IVCN of the same cycle. Climatology-persistence carries no intensity. Consensus aids and ECMWF tracks use climatological wind radii. "kt then" is the intensity at closest approach. Landfall is the first hour over land in the Natural Earth 50 m polygons.

A3. Ensemble summaries

	GEFS (18Z)	ECMWF (12Z)	AIFS (18Z)
Members	31	51	49
Within 100 / 200 / 300 km	0 / 0 / 6%	22 / 31 / 53%	31 / 53 / 78%
Median closest pass, km	717	288	173
Median peak intensity, kt	67	96	52
Tropical-storm wind in the city	0%	16%	4%
Landfall on Bahía de Banderas / Michoacán–Colima / Nayarit / Sinaloa	0 / 0 / 0 / 1	6 / 2 / 0 / 2	5 / 0 / 2 / 7
No landfall within the run	29	41	35
Median city rain, mm (all / 90th pct)	1 / 13	26 / 71	53 / 100

Companion to "Hurricane Simon and Puerto Vallarta: models, verification and uncertainty" (final report, 12 October 2026). Code and data: hurricane-simon/trudy in branch claude/hourly-forecast-animation-m4v1bj of the vallarta-weather repository. Not an official forecast. Follow the SMN, the NHC and Protección Civil. Emergencies: 911.